

SHREY SHARMA

AI/ML Engineer | Research Engineer — LLM Agents, Fine-Tuning, Evaluation

Bengaluru, India • +91-63969-39663 • shrey7shrey@gmail.com

github.com/ShreySharma07 • linkedin.com/in/shrey-sharma • huggingface.co/ShreySharma07

Summary

Computer Science graduate (AI specialization) building **multi-agent LLM systems, fine-tuning pipelines, and rigorous evaluation harnesses**. Designs agent architectures end to end — planning loops, tool-calling layers, RAG memory, and closed-loop verification — and reports results with measured baselines rather than anecdotes. Works in **Python, PyTorch, and Hugging Face Transformers**, with hands-on experience taking models from **QLoRA fine-tuning through Dockerized production deployment**. Seeking **AI/ML Engineer or Research Engineer** roles in applied LLM and agent research.

Technical Skills

- **LLM & Agents:** LLM Fine-Tuning, QLoRA, LoRA, PEFT, Quantization, RAG, Prompt Engineering, ReAct Agents, Multi-Agent Orchestration, Tool Calling, MCP, Model Evaluation & Benchmarking
- **ML/DL Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, TRL, Scikit-learn, ChromaDB, Google ADK
- **Languages:** Python, C++, SQL, Java
- **MLOps & Deployment:** Docker, FastAPI, AWS (SageMaker, Lambda, EC2, S3), MLflow, DVC, CI/CD, Git, Linux
- **Foundations:** Deep Learning, NLP, Linear Algebra, Probability & Statistics, Experiment Design, Data Structures & Algorithms

AI/ML Projects & Research

KARMA | Multi-Agent LLM System for Automated Code Optimization

[GitHub](#)

- Designed a **5-agent LLM orchestration pipeline** (classifier → planner → coder → verifier → reflector) that autonomously writes, tests, and iteratively refines GPU code, achieving **8.99× speedup** over the PyTorch eager baseline and **1.08×** over `torch.compile` on a consumer 6GB GPU — results prior work required H100 clusters to reach.
- Built a **closed-loop verification harness** (compile → 3-seed correctness check → determinism gate → interleaved benchmarking) that structurally blocks **LLM reward hacking**, yielding **0% false-positive correctness** across all reported results.
- Engineered a **RAG semantic memory layer (ChromaDB)** for cross-task knowledge transfer; a single authored knowledge entry lifted agent performance from **0.26× to 1.09×** on an identical task, demonstrating measurable in-context learning across runs.
- Running **ablation studies** (knowledge base, pre-flight profiling, reflection agent) on KernelBench toward a workshop paper submission.

Repliq | Autonomous AI Agent Platform for Workflow Automation

[GitHub](#)

- Architected a production-grade autonomous agent system that converts screen recordings into executable workflows using a **ReAct (Reason–Act–Observe) planning loop**, LLM reasoning, and iterative tool execution.
- Built an **MCP-compatible tool-calling layer** with full execution tracing, retry and error handling, and async orchestration via **FastAPI**, enabling reliable multi-step task automation.
- Engineered a **zero-trust security architecture** with Fernet-encrypted credential vaulting and **OAuth 2.0**, ensuring sandboxed agents never access long-lived secrets during autonomous execution.

Math-Specialized LLM Fine-Tuning | Llama 2, QLoRA, PEFT

[Hugging Face](#)

- Fine-tuned **Llama 2** on **GSM8K** for mathematical reasoning using **QLoRA, PEFT, and 8-bit quantization**, cutting GPU memory footprint by **70%** and reaching **0.83 training loss**.
- Shipped a **Dockerized inference pipeline** to Hugging Face Spaces, covering the full lifecycle from data preparation and training through evaluation and public deployment.

Education

B.Tech, Computer Science (AI Specialization) | University of Lucknow, Lucknow, India Nov 2022 – July 2026

- **Relevant Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Linear Algebra, Probability & Statistics, Data Structures & Algorithms

Achievements & Certifications

- **Top-5 Finalist, IIT Madras Malware Analysis Hackathon** — led a 4-person team building a deep learning malware classification model, placing top 5 of **120+ teams**.
- **Certifications:** Google Agentic AI (Kaggle/Google) • UC Berkeley Agentic AI MOOC • Machine Learning & Deep Learning (IBM/Coursera)